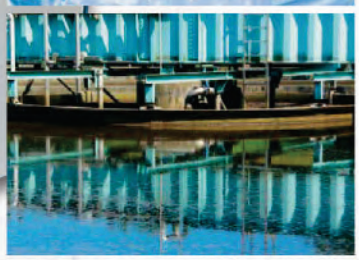
Alliance Instruments **Germany**

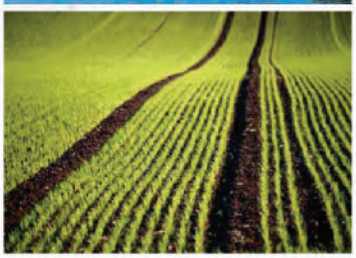
FLOWSYS

Continuous-Flow Analytik











ALLIANCE INSTRUMENTS

Segmented Continuous-Flow Analysis

CFA/SFA - Concept

For several decades Flow Analysis CFA technology provides a globally well established instrument concept. Samples can pass a series of complex analytical processing steps quickly in sequence, without influence to subsequent samples (no memory effect). Aspirated samples are splitted continuously to feed several analytical processes working simultaneously, so determining multiple analytes for each sample of a series. Typical sample throughputs of 30 samples per hour and four determinations simultaneously provides 120 analysis per hour for caibrants, samples and controls. The typical analytic method automated by CFA analysis combines sophisticated sample preparation with photometric detection.

CFA/SFA - Standards, Distinction

DIN EN ISO standards call technology 'Flow Analysis' and differentiate between CFA and FIA technique. Thereby CFA is the type of air-segmented Flow Analysis, whereas FIA is an unsegmented Flow Analysis type. First generation CFA was invented and established in the early 1960s and at this time a huge step in automation for chemical laboratories of clinic field and industry. FIA was developed in the late 1970s as an unsegmented alternative. After 30 years of worldwide use and a variety of valid APHA standard methods for CFA also European standards where developed and published starting in 1990s. Today accreditated laboratories can refer to a range of valid CFA-SFA standards for analytic methods in many different fields of application, as it is water/waste-water, agronomy-soil, food, tobacco. National regulations show growing acceptance for automated flow analysis due to valid EN ISO standards. In addition proven CFA methods exist for different fields of analytical chemistry, providing advantage with robustness, sample throughput, analysis costs and available LOD over some competing analytic methods using chromatography

CFA/SFA - Functional Structure

The segmented continuous flow analyser allways includes an automatic sampler for liquid samples, one or more CFA precision pumps with working reagents, one or more tray to carry the reaction system for each analytical procedure, a dual beam photometer (measure and reference) for each simultaneous determination with flowcell of adequate light path and finally the system control and data evaluation work place with PC and automatic calculating CFA software.

Working reagents for his CFA analyser are prepared by the user himself, following the method instruction of Alliance Instruments and using analytical quality laboratory chemical substances. Methods with integrated distillation additional require carrier gas (N2, laboratory air) and cooling water for stable condensation of the distillate.

CFA/SFA - Principle

The CFA multichannel precision pump always runs at same standard speed during analysis. In ready-standby mode all the working reagents and a 'zero-sample' are continuously pumped into the reaction system and this reaction solution is detected as true reagent blank (Baseline) in the flowcell. At the very begin of the reaction system air bubbles are add in regular intervals to split the reaction solution stream into regular liquid segments. With Microflow this segmentation appears about each second. This regular segmentation in CFA analysis minimise carryover of analyte concentration and causes intense mixing throughout the reaction system.

After the start of analysis mode sample and alternating zero-sample (wash, baseline) are aspirated in regular intervals for a series of calibrants, samples and controls. Running 30/hour each 120 seconds the next sample is aspirated. These 120s are typically used for 80s sample and 40s zero-sample wash. The air segmentation of CFA analysis splits each second this 80s sample stream. 80 sub-segments pass all the reagent dosing and reaction system and are detected in the flowcell, showing the CFA peak. Experienced users keep in mind, that segmented flow analysis do carry out repeated measurement on each sample and visualisis with the peak actual repeatability of the whole analytic process. Instrument operators use this benefit for quick quality checking, trouble shooting and for maintenance indication. Analysts appreciate to have the possibility of plausibility checking for each reasult during and after analysis. Segmentation air bubbles are removed before entering the flowcell, to assure pressure free reaction system.

Each sample series starts with calibrants measurment to set actual conditions. The CFA software automatically calculates the calibration function and with it the sample concentration.

Segmented CFA - Fields of Application

Today CFA analysis is typical routine technology for commercial laboratories, Institutes or work-groups with repeatedly larger sample series for anorganic analysis, for environmental science as water ecology, for marine research both in the laboratory and in site on the vessel. Large sewage plants use the cost efficient automation by CFA as well as the quality laboratory of industrial fertilizer production or the QC testing lab for tobacco production or legal limits. Complex sample matrix or analytic procedure steps or required special detection limits directs a user to apply CFA analysis for even small samples sets.

Routine testing in

Commercial lab

Disposal site regulation control for Cyanides and Phenols with integrated distillation, trace concentrations of Cyanides in potable water



Industrial

Quality Control

for example:

Legal quality and production control for Nicotine, sugar and Nitrates in tobacco extracts or smoke



Control of

Sewage

Determination of nitrogen and phosphorous as Nitrate, Nitrite, Ammonia, o.Phosphate, total-P, but also Chloride orSulphate



Aquatic and environm.

Systems Research

Analysis of marine chemistry for nutrients in the µg/L range, Oligotrophic seawater in the nanomol/L range



Agricultural Research Agric. Control

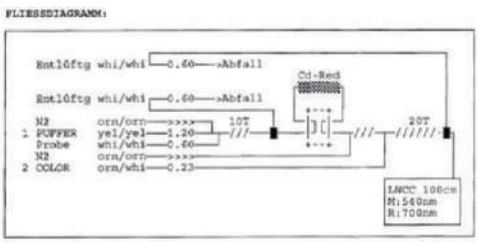
Typical analysis of Nmin for agricultural soils (NH4+NO3) P + K status in soils and for plant extracts

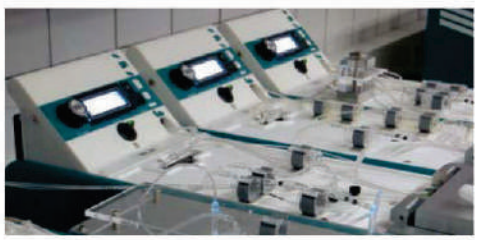


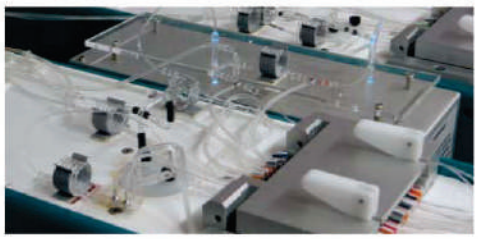
CFA-SFA
Analyzer FLOWSYS

- Each determination built on a separate Analytic Unit, working reagents inside the console
- Easy expandable and modifiable, flexible in use, reliable, maintenance and service-friendly design
- System control and automatic data evaluation software
- Electronic Unit with power supply and PC interface for connection of up to 6 (8) Analytic Units
- Automatic sampler, PC controlled
- Optional modules as UV-digestion, circulation cooling, detection by Flame Photometer (K, Na)











FLOWSYS - Analytic Unit

- Working reagents inside the console behind the front door
- Separate CFA pump per unit, air segmentation easy and free of noise
- Dimensions of reaction tray to build Microflow or Macroflow methods
- Reaction system built on leakage safe tray design with up to 2 h capacity
- Dual-beam photometer (measure and reference) for each Unit
- Electronic part of each Unit easy to remove in case of service
- Prepared for optional reagent valves

FLOWSYS - Reaction Tray (Manifold)

- The Flow Diagram in the method documentation describes the reaction system and helps the operator during instrument maintenance. Stipulated amounts of working reagents and sample require the use of specific CFA pump tubes quality
- The type, performance and concentration range of an analytic method is defined by the quantity (pump tube) of sample and each reagent and finally by the sequence of analytic procedure steps build by hardware as 'reaction system'

A variety of techniques are applied and built in sequence:

- Dosing of solutions
- Mixing and Dilution
- Provision of reaction and mixing time
- Flow-through inline dialysis for removal of disturbing sample matrix
- Acceleration or stabilisation of a reaction by incubation for certain time
- Hydrolysis under acid conditions near boiling point
- Digestion by oxidation under rigid UV-radiation and peroxodisulfat
- Selective digestion under specific UV radiation for complex cyanides

- Distillation of Cyanide (HCN) or steam-distillable Phenols according EN ISO standard to separate analyte from disturbing sample matrix

- Special CFA methods use the ability of segmented flow analysis to condition samples, to speparate the analyte in a two stage extraction procedure and after to measure the organic phase directly in the flowcell or to react the cleaned-up analyte and to measure the resulting colour.



FLOWSYS - Working reagents

- Working reagents are prepared by the operator, starting form analytical grade laboratory chemicals. The method documentation provides preparation instructions
- Minimised Reagent consumption and reagent waste with Microflow CFA
- Reagent bottles inside the console provide at least 8 -10 hours pumping quantity
- Shortest tubing distance for reagent aspiration on the Flowsys





FLOWSYS - Detection

- At the end of the reaction system the reaction solution is measured continuously in a flow cell, detection wavelength as specified for the method
- The Flowsys standard photometer uses dual beam compensation between measure and reference detection. Photometer light souce by specific LED's
- Matrix Photometers with bichromatic blanking are used for detection of solutions containing high levels of salt, especially sea water. Optical interference is compensated as well as disturbing micro-bubbles
- Depending on the intensity range of colour produced in the reaction solution, flow cells of 15mm to 50mm light path are applied for detection
- LWCC detectors with 1000mm light path are available and have been successfully implemented for nanomolar detections in marine research
- Other detectors with continuous measurement of intensity can be integrated with the CFA analyser for simultaneous determination and automatic evaluation. This regards mostly Flame Photometers or AAS, in specific case it can be an electrode



FLOWSYS - Electronic Unit

- Power supply for up to 6 (8) Analytic Units
- PC Interface (ADC) for up to 6 (8) Analytic Units
- Optional timer interface for automatic switching and shut down



FLOWSYS - Sampler

- Standard sampler XY-240, compact and efficient
- 4 sample racks with each 5x12 positions (60) or 6x15 positions
- Sample cups of 1.5ml to 50ml volume can be configured
- Save aspiration of the zero-sample (Baseline)

Alliance Instruments

Alliance Instruments GmbH develops and distributes analyzers for automation of wetchemical-colorimetric analysis procedures. Main business focus since founding in 1990 is on segmented Continuous-Flow Analysis technology (CFA - SFA).
Related technology as Sequential Automatic Analysis (SAA, discrete analysis) alternatively may provide specific advantage in certain fields of use, depending on customers laboratory situation.

Production

The Alliance Instruments network of CFA specialists provided sufficient ressources during the 1990s, to build suitable CFA-SFA solutions for our customers, starting from basic instrumentation modules. Growing demand for particular performance of analytic preparation and reaction systems soon initiated our own developments and production of optimised modules and technology. Since 2009 Alliance Instruments GmbH have consolidated its assembling and production of CFA-SFA systems in our German branch. SAA analysis have been installed since 1998 for well defined fields of application, using our knowledge for method implementation.

Consulting

Customers in several European countries use our technology and instrumentation in daily routine for a variety of applications and analytic targets. Different branches, tasks and working ranges often require customised solutions, to gain best success for the investment into new automation technology. CFA-SFA technology may help the customer to replace manual sample preparation procedures by an automated CFA procedure, providing adequate function during the method test sequence and to save laboratory ressources with it. Correct results with colorimetric methods for inorganic analysis or convention parameters may require additional measures to be taken, for matching individual need of sample pH or other sample matrix and so to avoid interference. Alliance Instruments appreciates the thorough discussion of customers analytical targets and samples nature in advance and provide to new or existing customers of CFA-SFA technology sufficient expertise for it.

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